

INFO8006 Introduction to Artificial Intelligence

Exercise session 3: Quantifying uncertainty

RECAP

Kolmogorov's axioms

A *probability space* is a *sample space* Ω , whose elements ω are the *sample points*, equipped with a probability function P such that $0 \leq P(\omega) \leq 1$ and $\sum_{\omega \in \Omega} P(\omega) = 1$. An *event* is a set of sample points $A \subseteq \Omega$, of probability $P(A) = \sum_{\omega \in A} P(\omega)$. Hence $P(\Omega) = 1$, $P(A \cup B) = P(A) + P(B)$ for disjoint events, $P(\neg a) = 1 - P(a)$ and $P(a \vee b) = P(a) + P(b) - P(a \wedge b)$.

Random variables

A *random variable* is a function $X : \Omega \rightarrow D_X$, with $P(X = x) = \sum_{\{\omega : X(\omega) = x\}} P(\omega)$. A number is written $P(x)$, short for $P(X = x)$; a whole *distribution* is written $\mathbf{P}(X)$, *i.e.* the table of the values $P(X = x)$ for all $x \in D_X$.

Conditional probability, sum and product rules

For $P(b) > 0$, the *conditional probability* of a given b is $P(a | b) = P(a, b)/P(b)$, hence the *product rule* $P(a, b) = P(a | b)P(b) = P(b | a)P(a)$. Summing a joint distribution over the values of a variable marginalizes it out (*sum rule*), $P(x) = \sum_y P(x, y)$, and with the product rule it gives the *law of total probability*, $P(x) = \sum_y P(x | y)P(y)$. Selecting the entries consistent with the evidence $\mathbf{e}$, marginalizing the hidden variables out and normalizing gives the posterior over a query variable Q , $\mathbf{P}(Q | \mathbf{e}) = \frac{1}{Z} \mathbf{P}(Q, \mathbf{e})$ with $Z = \sum_q P(q, \mathbf{e})$, the constant $1/Z$ being also written α .

Bayes' rule

From the two factorizations above,

$$P(a | b) = \frac{P(b | a)P(a)}{P(b)},$$

where $P(a)$ is the prior belief on a , $P(b)$ the probability of the evidence b , $P(a | b)$ the posterior belief on a given b , and $P(b | a)$ the likelihood.

Chain rule

Any joint distribution is an incremental product of conditional distributions, $P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1, \dots, x_{i-1})$.

Independence

$A \perp B$ iff $P(a, b) = P(a)P(b)$, and $A \perp B | C$ iff $P(a, b | c) = P(a | c)P(b | c)$, for all values of the variables. Conditional independence factors a joint distribution into smaller tables, which is what makes probabilistic models scale.

In session: exercises 1, 2, 3, 5.

Exercise 1 Rational beliefs (AIMA ch. 12)

Let A and B be two *events* over the probability space Ω . An agent holds the beliefs $P(A) = 0.4$ and $P(B) = 0.3$. What ranges of probabilities would it be rational for the agent to hold for the events $A \cup B$ and $A \cap B$? What if the agent also believes that $P(A | B) = 0.5$?

Solution.

1. The agent's beliefs should satisfy

$$0.4 = \max\{P(A), P(B)\} \leq P(A \cup B) \leq \min\{1, P(A) + P(B)\} = 0.7$$

$$0 = \max\{0, P(A) + P(B) - 1\} \leq P(A \cap B) \leq \min\{P(A), P(B)\} = 0.3.$$

This can be illustrated with Venn diagrams.

2. If the agent believes that $P(A | B) = 0.5$, it should also believe that

$$P(A \cap B) = P(A | B)P(B) = 0.5 \times 0.3 = 0.15,$$

by the product rule. Then, the agent should rationally believe that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.3 - 0.15 = 0.55.$$

Exercise 2 Full joint distribution (AIMA ch. 12)

A monitoring program supervises the bearings of a wind turbine. A worn bearing (*fault*) tends to make the turbine vibrate (*vibration*) and to trigger the alarm of a diagnostic probe (*alarm*). The program holds the joint probability table below. Compute the following probabilities.

	vibration		no vibration	
	alarm	no alarm	alarm	no alarm
fault	0.108	0.012	0.072	0.008
no fault	0.016	0.064	0.144	0.576

Solution. In the following, we represent the events *vibration*, *alarm* and *fault* by the random variables $V, A, F : \Omega \rightarrow \{0, 1\}$, respectively. The table above is then the joint probability distribution $\mathbf{P}(V, A, F)$. When clear from context, we abbreviate $P(V = v, A = a, F = f)$ as $P(v, a, f)$.

- (a) $P(\text{vibration})$

Solution.

$$P(V = 1) = \sum_a \sum_f P(V = 1, a, f) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$

- (b) $P(\text{fault})$

Solution.

$$P(F = 1) = \sum_v \sum_a P(v, a, F = 1) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2$$

(c) $P(\text{vibration} \mid \text{fault})$

Solution.

$$\begin{aligned} P(V = 1 \mid F = 1) &= \frac{P(V = 1, F = 1)}{P(F = 1)} = \frac{\sum_a P(V = 1, a, F = 1)}{P(F = 1)} \\ &= \frac{0.108 + 0.012}{0.2} = 0.6 \end{aligned}$$

(d) $P(\text{fault} \mid \text{vibration} \cup \text{alarm})$

Solution.

$$\begin{aligned} P(F = 1 \mid V = 1 \cup A = 1) &= \frac{P(V = 1 \cup A = 1, F = 1)}{P(V = 1 \cup A = 1)} \\ &= \frac{P(F = 1) - P(V = 0, A = 0, F = 1)}{1 - P(V = 0, A = 0)} \\ &= \frac{0.2 - 0.008}{1 - 0.584} = 0.4615 \end{aligned}$$

Exercise 3 Fraud detection (AIMA ch. 12)

A fraud detection program screens card payments. It is 99% accurate, *i.e.* it flags a fraudulent payment with probability 0.99 and a legitimate payment with probability 0.01. Fraud is rare: it strikes only 1 payment in 10 000. The program has just flagged your payment. Why is it good news, for you, that fraud is rare? What is the probability that the payment is actually fraudulent?

Solution. Let f be the event that the payment is fraudulent and a the event that it is flagged. Rarity is good news because the prior $P(f)$ is small: most flagged payments are legitimate payments flagged by mistake. Indeed,

$$\begin{aligned} P(f \mid a) &= \frac{P(a \mid f)P(f)}{P(a)} \\ &= \frac{P(a \mid f)P(f)}{P(a \mid f)P(f) + P(a \mid \neg f)P(\neg f)} \\ &= \frac{0.99 \times 10^{-4}}{0.99 \times 10^{-4} + 0.01 \times (1 - 10^{-4})} = 0.0098. \end{aligned}$$

Exercise 4 Conditional independence (AIMA ch. 12)

For each of the following statements, either prove it is valid or give a counterexample.

(a) If $P(a, b, c) > 0$ and $P(a \mid b, c) = P(b \mid a, c)$, then $P(a \mid c) = P(b \mid c)$.

Solution. True. By the product rule, $P(a \mid b, c)P(b, c) = P(a, b, c) = P(b \mid a, c)P(a, c)$. Since $P(a, b, c) > 0$, $P(a \mid b, c) = P(b \mid a, c) > 0$, and dividing by it gives $P(b, c) = P(a, c)$ and thereby $P(b \mid c) = P(a \mid c)$.

(b) If $P(a \mid b, c) = P(a)$, then $P(b \mid c) = P(b)$.

Solution. False. The statement $P(a \mid b, c) = P(a)$ merely states that a is independent of b and c , it makes no claim regarding the dependence of b and c . Counterexample: a and b record the results of two independent coin flips, and $c = b$.

(c) If $P(a \mid b) = P(a)$, then $P(a \mid b, c) = P(a \mid c)$.

Solution. False. While the statement $P(a \mid b) = P(a)$ implies that a is independent of b , it does not imply that a is conditionally independent of b given c . Counterexample: a and b record the results of two independent coin flips, and $c = a \vee b$.

Exercise 5 Bag of coins (AIMA ch. 13)

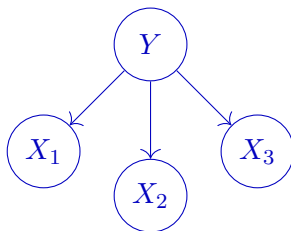
We have a bag of three biased coins a , b , and c with probabilities of coming up heads of 20 %, 60 %, and 80 %, respectively. One coin is drawn randomly from the bag (with equal probability of drawing each of the three coins), and then the coin is flipped three times to generate the outcomes X_1 , X_2 and X_3 (heads or tails).

- (a) Draw the Bayesian network corresponding to this setup and define the corresponding conditional probability tables (CPTs).

Solution. The process is described with the following random variables:

- $Y \in \{a, b, c\}$, the coin drawn from the bag.
- $X_i \in \{h, t\}$, the realization of the i -th coin toss.

The variable Y is the parent of X_i , it should be put before in the network ordering.



Y	$P(Y)$	$P(X_i = h \mid Y)$
a	$\frac{1}{3}$	0.2
b	$\frac{1}{3}$	0.6
c	$\frac{1}{3}$	0.8

- (b) Determine which coin (a , b or c) is most likely to have been drawn from the bag if the observed tosses came out heads twice and tails once.

Solution. In order to find the most likely coin, we have to calculate the conditional probability distribution $P(Y \mid X_1 = h, X_2 = h, X_3 = t)$ ¹. By Bayes' rule, we have

$$P(Y \mid X_1 = h, X_2 = h, X_3 = t) = \alpha P(X_1 = h, X_2 = h, X_3 = t \mid Y)P(Y).$$

But we know that X_1 , X_2 and X_3 are independent with respect to Y , meaning that

$$P(X_1, X_2, X_3 \mid Y) = P(X_1 \mid Y)P(X_2 \mid Y)P(X_3 \mid Y).$$

Then, we have

$$P(Y \mid X_1 = h, X_2 = h, X_3 = t) = \alpha \times \frac{1}{3} \times \begin{cases} 0.2 \times 0.2 \times 0.8 = 0.032 & \text{if } Y = a \\ 0.6 \times 0.6 \times 0.4 = 0.144 & \text{if } Y = b \\ 0.8 \times 0.8 \times 0.2 = 0.128 & \text{if } Y = c \end{cases}$$

meaning that b is the most likely coin to have been drawn. Interestingly, we don't need to normalize the distribution as we only care about the maximum.

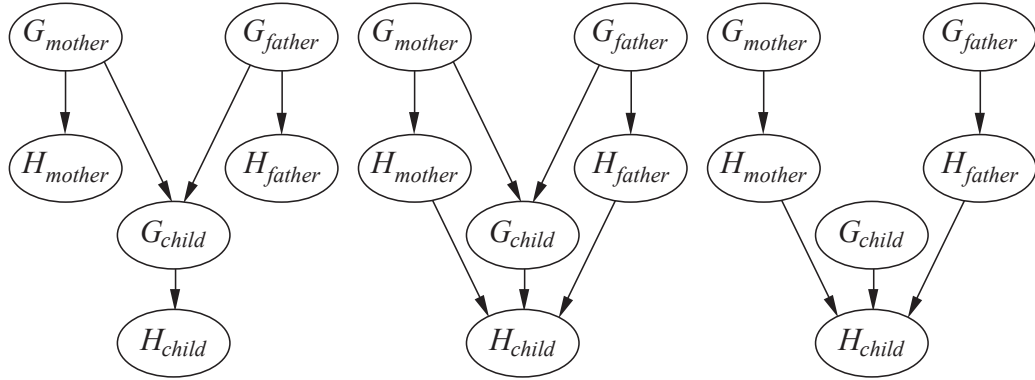
Exercise 6 Handedness (AIMA ch. 13)

Let H_x be a random variable denoting the handedness of an individual x , with possible values l or r . A common *hypothesis* is that left or right-handedness is inherited by a simple mechanism; that is, there is a gene G_x , also with values l or r , and H_x turns out the same as G_x with some probability s .

Furthermore, the gene itself is equally likely to be inherited from either of the individual's parents, with a small nonzero probability m of a random mutation flipping the gene.

¹It should be noted that the order of the tosses is irrelevant as they are performed independently.

- (a) Which of the following networks claim that $P(G_{\text{child}} | G_{\text{mother}}, G_{\text{father}}) = P(G_{\text{child}})$?



Solution. The third one, as the node G_{child} does not have any parents.

- (b) Which of the networks make independence claims that are consistent with the hypothesis about the inheritance of handedness?

Solution. The first and second ones. It should be noted that the second one is also valid as it makes *fewer* assumptions about the dependencies between the variables, *i.e.* it is more general than the first one.

- (c) Write down the CPT for the G_{child} nodes in first network, in terms of s and m .

Solution.

G_{mother}	G_{father}	$P(G_{\text{child}} = l G_{\text{mother}}, G_{\text{father}})$
l	l	$1 - m$
l	r	0.5
r	l	0.5
r	r	m

- (d) Suppose that $P(G_{\text{mother}} = l) = P(G_{\text{father}} = l) = q$. In the first network, derive an expression for $P(G_{\text{child}} = l)$ in terms of s , m and q .

Solution.

$$\begin{aligned}
 P(G_{\text{child}} = l) &= \sum_{g_m} \sum_{g_f} P(G_{\text{child}} = l, g_m, g_f) \\
 &= \sum_{g_m} \sum_{g_f} P(G_{\text{child}} = l | g_m, g_f) P(g_m) P(g_f) \\
 &= (1 - m)q^2 + 0.5(1 - q)q + 0.5q(1 - q) + m(1 - q)^2 \\
 &= q + m - 2qm
 \end{aligned}$$

- (e) Under conditions of genetic equilibrium, we expect the distribution of genes to be the same across generations. Use this to calculate the value of q , and, given what you know about handedness in humans, explain why the hypothesis described at the beginning of this question must be wrong.

Solution. Under genetic equilibrium, we have $P(G_{\text{child}} = l) = P(G_{\text{parent}} = l) = q$. Therefore, $q = q + m - 2qm$ meaning that $q = 0.5$. In practice, being left-handed is much rarer than being right-handed, which invalidates the initial hypothesis.

1. The Bayes' rule states that ...

- ☐ $P(x | y) = \frac{P(y | x)P(y)}{P(x)}.$
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- ☐ $P(y | x) = \frac{P(x | y)P(x)}{P(y)}.$
- ☐ $P(x) = \frac{P(x | y)P(y)}{P(x)}.$

2. Which of the following is true?

- ☐ $P(x, y | z) = \frac{P(x | y, z)}{P(y, z | x)}.$
- ☐ $P(x, y, z) = P(x)P(y)P(z).$
- ☒ $P(x, y | z) = P(y | x, z)P(x | z).$
- ☐ $P(x | y, z) = \frac{P(x, y, z)}{P(y)P(z)}.$

3. Naive Bayes model assumes ...

- ☒ conditional independence between the effects, given the cause.
- ☐ independence between the effects.
- ☐ independence between the effects and the cause.
- ☐ conditional independence between each effect and the cause, given the other effects.

4. The Bayes' rule can be read as "The posterior is equal to ...

- ☐ the product of the likelihood and the prior".
- ☐ the ratio between the posterior distribution and the evidence, times the prior".
- ☐ the likelihood to prior ratio multiplied by the evidence".
- ☒ the likelihood to evidence ratio multiplied by the prior".